KOR-PE-01 Critical Hardware/Environmental Fault Resolution SOP (Cisco CLI)

Node: KOR-PE-01 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_8951

Description

The failure domain of ASIC memory parity errors, line card crashes, and power supply redundancy loss on the PE role in the Cisco vendor equipment at the Koramangala Datacenter is primarily attributed to the inherent vulnerability of the custom ASIC memory modules to environmental stressors such as high-temperature fluctuations and humidity. Furthermore, the absence of adequate redundancy in the power supply domain exacerbates the likelihood of these faults occurring, as any single point of failure can precipitate a cascade of subsequent errors. Moreover, the Cisco IOS XR software's reliance on the ASIC memory for its control plane operations intensifies the severity of these faults, necessitating prompt and accurate diagnosis to prevent network disruptions.

Pre-requisite Checks

- Verify the site location and ensure that the Koramangala Datacenter is accessible.

- Check the redundant link status to ensure that the secondary link is operational.

- Dispatch remote hands to the Koramangala Datacenter to facilitate physical access to the affected equipment.

- Confirm that the power supplies are functioning within the specified operating range and that the ambient temperature is within the recommended limits.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to high-temperature fluctuations	High
Line card crashes due to power supply redundancy loss	Medium
Environmental stressors (humidity, dust, etc.) affecting the custom ASIC memory modules	Low

Diagnostic Commands & Expected Outputs

```
Command: show hw-module power-supply, Expected Output: Power supply information, including the current status and operating range of the power supplies., Description: This command isolates the power supply domain and verifies its operational status, ensuring that the power supplies are functioning within the specified operating range.

Command: show hw-module temperature, Expected Output: Temperature information, including the current temperature and thermal thresholds., Description: This command isolates the environmental stressors affecting the custom ASIC memory modules, confirming that the ambient temperature is within the recommended limits.
```

```
Command: show hw-module asic-memory, Expected Output: ASIC memory information, including the current status and parity error counts., Description: This command isolates the ASIC memory domain and verifies its operational status, detecting any parity errors that may indicate a hardware fault.
```

Resolution Steps

Step 1: Step: Replace the faulty power supply unit (PSU), Command: hw-module power-supply replace , Description: This step replaces the faulty PSU with a new one, restoring power supply redundancy and mitigating the risk of line card crashes.

Step 2: Step: Swap the line card with a spare one, Command: hw-module line-card replace , Description: This step swaps the faulty line card with a spare one, restoring control plane operations and preventing network disruptions.

Step 3: Step: Perform a hardware reset on the affected node, Command: hw-module reset , Description: This step performs a hardware reset on the affected node, clearing any accumulated errors and ensuring a clean restart of the control plane.

Post-Check Verification

```
Command: show hw-module power-supply, Expected Output: Power supply information, including the current status and operating range of the power supplies., Description: This command verifies that the power supplies are functioning within the specified operating range and that the power supply redundancy is restored.

Command: show hw-module temperature, Expected Output: Temperature information, including the current temperature and thermal thresholds., Description: This command verifies that the ambient temperature is within the recommended limits and that the environmental stressors are mitigated.

Command: show hw-module asic-memory, Expected Output: ASIC memory information, including the current status and parity error counts., Description: This command verifies that the ASIC memory is operational and that parity errors are no longer detected.
```

Roll-back Procedures

- Step: Restore the original power supply unit (PSU), Command: hw-module power-supply restore , Description: This step restores the original PSU, reverting to the previous power supply configuration and mitigating any potential disruptions.

- Step: Replace the line card with the faulty original one, Command: hw-module line-card replace , Description: This step replaces the line card with the faulty original one, reverting to the previous control plane configuration and restoring network operations.

- Step: Perform a hardware reset on the affected node, Command: hw-module reset , Description: This step performs a hardware reset on the affected node, clearing any accumulated errors and ensuring a clean restart of the control plane.